

McHenry “Mack” Fisher

DoW Secret Clearance

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Education

The Pennsylvania State University

Mechanical Engineering B.S. (Expected May 2029) - GPA 3.93

University Park, PA
August 2025—Present

Engineering Experience

The Applied Research Laboratory at Penn State University

R&D Engineer Intern, Advanced Materials Processing | MMSO

State College, PA
October 2025—Present

- Led end-to-end development of a cold-spray robot for repair of cylindrical metal components; integrated linear and rotary motion, a swivel mount, closed-loop stepper motors, drivers, and limit switches
- Designed and advanced a bellows-sealed enclosure concept into product development after a 3-day team design sprint, enabling nozzle travel while containing powder around a rotating component
- Created approximately 10 manufacturing drawings and machined robotic mounts, specifying perpendicularity and clearance fits for locating bolts; coordinated welded fabrication with a machinist
- Quantified coating porosity through metallographic sample preparation and Keyence microscopy
- Specified hardware, tolerances, and fits; procured components through McMaster-Carr and SendCutSend
- Reviewed spark plasma sintering literature on applications, processing parameters, and material behavior
- Automated OrcaSlicer-to-GRBL G-code conversion using an AI-assisted Python script, saving approximately 15 minutes of manual processing per file
- Supported V2 funding approval by demonstrating coordinated linear and rotary raster motion in a compressed-air dry run and sharing test videos with prospective users

Nittany Motorsports Club

Chassis and Suspension Subsystems Engineer

University Park, PA
October 2025—November 2025

- Reduced titanium suspension end-plate weight using Fusion 360 Generative Design to improve strength-to-weight performance
- Designed a tubular-chassis torsional-rigidity test fixture in CAD and began developing its loading and measurement procedure

Professional Experience

Top Shop Auto

Automotive Technician Intern

Walnut Creek, CA
January 2025—April 2025

- Modified a Camaro front end, and chassis for a different-generation appearance; installed a wiring harness, sourced subchassis parts and panels, and supported radiator-mount modifications
- Designed vehicle components in Fusion 360 and fabricated approximately 6 installed parts, including 3D-printed switch mounts, gauge clusters, and a sheet-metal battery box
- Diagnosed vehicle faults using scan tools, compression testers, and multimeters; assisted with flathead V8 disassembly and operated shop lifting and fabrication equipment

Honors & Awards

Dean's List, Fall 2025 & Spring 2026 · Campolindo Technical Excellence Award, 2025 · AP Scholar with Distinction · Football Team Captain, Campolindo HS, 2025

Leadership

Engineering Orientation Network (EON) | Mentor

Led 40 first-year engineering students through orientation and continue mentoring them on career opportunities and campus involvement through a group chat

August 2026—Present

Projects

1964 Impala Engine Swap (Client Project)

April 2025—August 2025

- Completed a Chevy 350 engine swap within a five-month schedule and budget; transferred the intake manifold and accessory drives
- Commissioned the replacement powertrain through carburetor tuning, ignition timing, wiring repairs, and fuel-system diagnostics using a multimeter, fuel-pressure gauge, and timing light
- Delivered a running client vehicle while managing parts sourcing, delays, receipts, and progress updates; identified incorrect fuel-pump placement and recommended professional relocation

1974 Honda CB550 Resto-Mod “Café Racer”

April 2023—January 2025

- Restored a 1974 Honda CB550 through an independent frame-off rebuild; completed mechanical, electrical, welding, and fabrication work, with the motorcycle passing a mechanic’s inspection and remaining in regular use
- Fabricated and welded a seat delete and U-bar for a café racer conversion and formed a sheet-metal battery box to house modern electronics
- Overhauled drivetrain, suspension, and braking systems by replacing bearings and seals, fitting spoked wheels, and installing performance brake upgrades
- Integrated custom wiring, LED lighting and diodes, an upgraded regulator-rectifier, and replacement ignition; configured timing and resolved electrical integration issues

Arduino Dog Feeder

January 2024—February 2024

- Built a working auger-based feeder using Fusion 360 and 3D printing; selected a NEMA stepper motor and TMC-series driver based on torque requirements
- Eliminated dispensing jams through 3 CAD and 3D-printed design iterations that corrected clearance between the ramp and auger
- Programmed Arduino control of button-activated dispensing, motor speed, rotation count, duration, and direction; verified operation through bench testing

Sigma Pi DJ Stand

August 2026

- Delivered a custom DJ stand in 4 days by independently designing, sourcing, and fabricating it for Sigma Pi; it remains in use approximately 3 times per week
- Designed an approximately 11 × 5 × 5 ft stand as 6 detachable sections using CAD, dimensioned sketches, and cut lists, enabling assembly in about 2 minutes and transport by 1–2 people per section

Wind-Turbine Gearbox Mount

October 2025

- Designed a mount in Fusion 360 for PSU Wind Energy Club to integrate a planetary gearbox inline with the wind-turbine shaft
- 3D printed the mounting assembly and verified fit and operation through installation and testing on the turbine

Technical Skills

CAD & Product Development: SolidWorks (Parts, Assemblies, Configurations, Exploded Views, Drawings), Fusion 360, CAD/CAM, Design for Manufacturing & Assembly (DFM/DFA), GD&T, Tolerance Stack-Ups, Bills of Materials (BOMs), Component Selection

Manufacturing: CNC Process Planning, Toolpaths, Feeds & Speeds, Workholding, CNC/Manual Milling, Lathe, Waterjet/Laser/Plasma Cutting, Press Brake, MIG/TIG Welding, 3D Printing

Metrology & Testing: Calipers, Micrometers, Height Gauges, Dial Indicators, Test Requirements, Data Collection, Validation, Systematic Diagnostics

Robotics & Controls: Motion Control, Servo/Stepper Motors, Encoders, Limit Switches, Homing, Sensors, Relays, Motor Controllers

Electronics & Documentation: Multimeter, Oscilloscope, Power Supplies, Breadboarding, Wiring, Soldering, Design Reviews, Work Instructions, Assembly Procedures, Revision Control

Programming & Materials: MATLAB (Calculations, Plotting), Arduino, Python (Working Knowledge), G-code/UGS, Cold Spray, Metallography, Porosity Analysis, Spark Plasma Sintering (SPS)